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BOILING SENSING BASED ON ACOUSTIC RECOGNITION AND DEEP LEARNING

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ABSTRACT

Anomaly detection in nuclear power plants enables us to early execute prevention and mitigation measures against accident progression. In sodium-cooled fast reactors, coolant boiling in reactor cores is one of the important phenomena in the safety assessment. Our final target of the present study is to realize the acoustic anomaly detection of the boiling inception in actual reactors. In the actual environment, various sorts of noises are expectedly superposed on accidental boiling sounds. It is inevitable to distinguish the boiling sounds from the superimposing hostile disturbance with high accuracy. To achieve this, we utilize machine learning techniques and assess the feasibility of boiling sensing based on acoustic recognition and deep learning. In the present study, we employ an autoencoder to denoise boiling sounds, and a convolutional neural network to detect the boiling inception. The boiling acoustics have not been fully understood yet. In the present study, we find that some characteristics of the boiling acoustics are consistent with the resonance vibration of the heating body. This finding contributes to elucidating the physics of boiling acoustics. In addition, it helps us detect boiling occurrences with high accuracy judging from the acoustic characteristics' patterns.

Keywords: Boiling, Boiling Acoustics, Acoustic detection, Machine learning, Deep neural network

NOMENCLATURE

l	wire length
d	wire diameter
ρ	density of Pt wire
E	Young's modulus
λ	mode parameter of resonance
V	voltage
I	electric current
q	heat flux

1. INTRODUCTION

Anomaly detection in nuclear power plants enables us to early execute prevention and mitigation measures against accident progression. In sodium-cooled fast reactors (SFR), coolant boiling in reactor cores is one of the important phenomena in the safety assessment (e.g. [1]). It will induce a meltdown of a fuel assembly. Its symbolic example is the Fermi-I partial-meltdown accident [2]. The anomaly detection of the coolant boiling can help us prevent the start and progress of the meltdown accident.

Optical measurements to detect states of boiling phenomena have been investigated and developed so far. One of the latest studies is the infrared thermometry detection of boiling crisis together with the deep learning method [3]. However, in terms of application to nuclear power plants, optical measurement techniques are not suitable. An acoustic measurement is a robust, noise-resistant, and remote-sensing technique that can work even under severe conditions in nuclear power plants. Recently, acoustic identification with deep learning was utilized for the prediction of the boiling crisis [4,5], inception, and state transition [6].

Our final target of the present study is to realize the acoustic anomaly detection of the boiling inception in actual reactors. In the actual environment, various sorts of noises are expectedly superposed on accidental boiling sounds. It is inevitable to

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distinguish the boiling sounds from the superimposing hostile disturbance with high accuracy. To achieve this, we utilize machine learning techniques and assess the feasibility of boiling sensing based on acoustic recognition and deep learning. In the present study, we employ an autoencoder (AE) to denoise boiling sounds, and a convolutional neural network (CNN) to detect the boiling inception.

The boiling acoustics have been intensively investigated and experimentally characterized so far [4-12]. It is generally accepted that the boiling sound originates from vapor bubble generation, growth, and condensation. Yet, a comprehensive theory of boiling acoustics has not been obtained. Toward it, we investigate correlations between solid-structure vibrations and boiling acoustics. It will contribute to effective and reliable feature extraction of the boiling sound.

2. DATA ACQUISITION OF BOILING SOUND

The present experimental system is that a small-diameter platinum wire is horizontally fixed and electrically heated in a distilled water container (see FIGURE 1). The water container is 180 mm in diameter and 280 mm in height. The platinum wire is 40 mm in length and 300 μm in diameter. The purity is 99.95%. A bulk temperature of the distilled water is maintained at approximately 80°C. A microphone (Sennheiser MKE600) is located near the container and acquires sound data with and without subcooled boiling. Deaeration of the distilled water is performed by 30-minute boiling before the sound recording. Electric current through the platinum wire is evaluated from the voltage of shunt resistance. The applied voltage on the platinum wire is measured by a digital multiplexer. The wire temperature is evaluated by its electric resistance. The heat flux of the wire is evaluated from the following relationship:

$$q = \frac{VI}{2\pi dl} \tag{1}$$

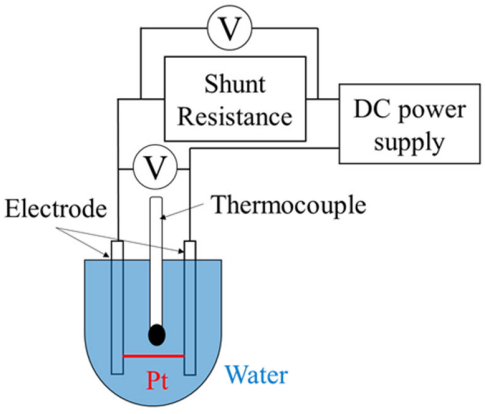
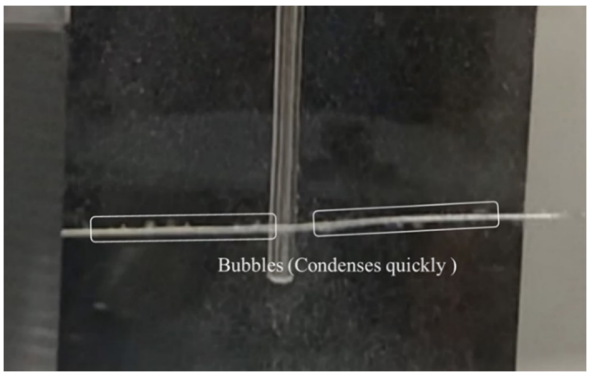


FIGURE 1: POOL BOILING EXPERIMENTAL APPARATUS.

FIGURE 2 shows photographs of boiling bubbles in the case of subcooled and saturated boiling. In the case of subcooled boiling, small bubbles are repeatedly formed and then condensate quickly. On the other hand, in the case of saturated boiling, we observe that relatively large bubbles are formed and then depart from the wire.



(a) SUBCOOLED BOILING.



(b) SATURATED BOILING.

FIGURE 2: PHOTOGRAPH OF BOILING BUBBLES.

3. BOILING ACOUSTICS

We employ the discrete Fourier transform to obtain acoustic spectrums of the boiling. To further understand the boiling acoustics, we compare the spectrums with a resonance frequency of a circular pillar. We assume that the circular pillar is fixed on both ends. The resonance frequency is given as:

$$f_i = \frac{(\lambda_i l)^2}{8\pi l^2} d \sqrt{\frac{E}{\rho}} \tag{2}$$

The mode parameter of the resonance is shown in TABLE 1. We evaluate values of the Pt density and Young's modulus [13] with consideration of temperature dependency. The wire expands in the longitudinal direction as well when it is heated. Because of this, we assume that the wire length ranges from 40.1 to 41.0 mm in the present evaluation. FIGURE 3 shows examples of the acoustic spectrums of the boiling concerning the resonance frequency. Blue broken lines correspond to the resonance

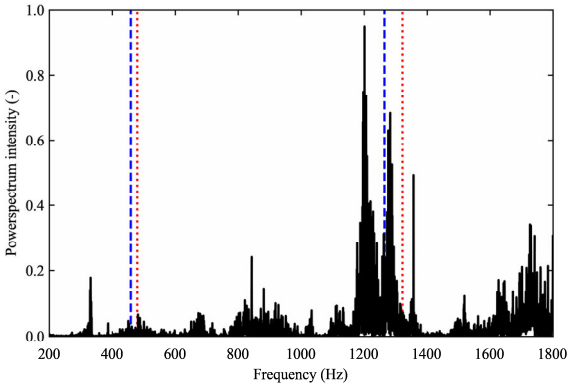
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frequency in the case of the longest wire length, and red ones do that in the case of the shortest length. The probable resonance frequency is located between them.

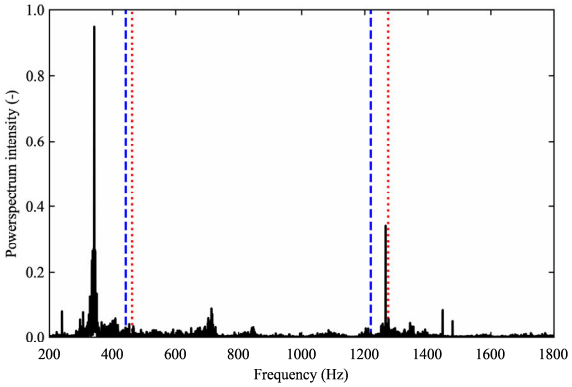
In both CASE 1 and 2, we find that one of the spectrum peaks is located on the resonance frequency which ranges from approximately 1200 to 1400 Hz. It indicates a part of the boiling acoustics is characterized by the resonance of the heating body. We speculate that other spectrum peaks can be related to the vapor bubble formation, deformation, and condensation. In CASE 3, one spectrum peak is located around 500 Hz. It is consistent with the previous study [6]. The peak is located on the first resonance frequency. In this case, a significant peak cannot be found in the range of 1200 to 1400 Hz. On the other hand, the peak is located around 2500 Hz, where the third resonance frequency is expected. The resonance modes are shifted in different cases. The main source of vibration excitation is expected to be boiling bubble dynamics. Bubble nucleation can differ on different surface conditions. It may cause a change in the vibration excitation.

TABLE 1: MODE PARAMETER OF RESONANCE.

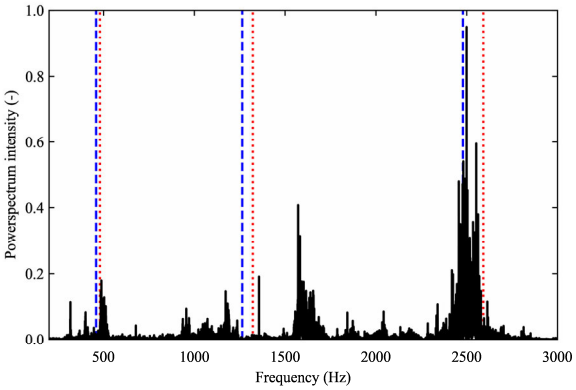
$\lambda_1 l$	4.730
$\lambda_2 l$	7.853
$\lambda_3 l$	10.996
$\lambda_4 l$	14.137



(a) CASE 1.



(b) CASE 2.



(c) CASE 3.

FIGURE 3: ACOUSTIC SPECTRUMS OF BOILING AND RESONANCE FREQUENCY.

4. DEEP LEARNING OF BOILIG SOUND

The train and test data we use are the pool boiling acoustic spectrums, described above for proof of concept of the acoustic boiling sensing. In the actual environment, various sorts of noises are expectedly superposed on accidental boiling sounds. In the present study, we superposed white noise on the boiling sounds, which corresponds to hostile disturbance against machine learning. FIGURE 4 illustrates an example of boiling acoustic spectrums with different intensities of white noises. The white noise intensity is quantified as the noise-to-signal (N/S) ratio. Note that the characteristic spectrum peaks become indistinguishable from the white noise, as the noise becomes intense.

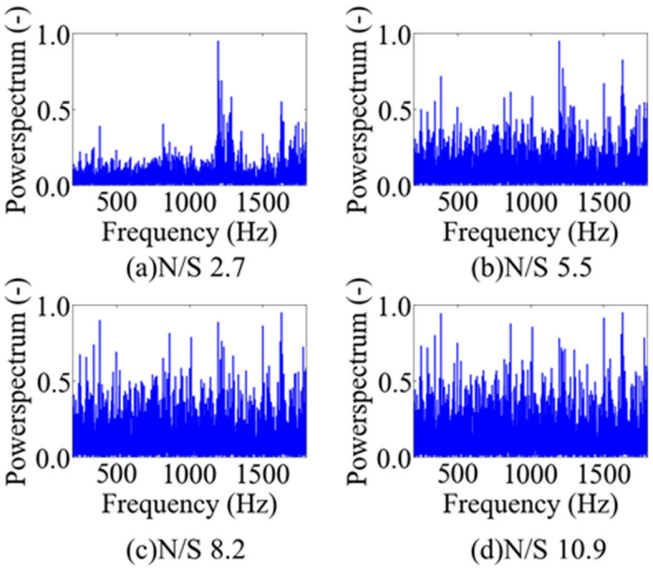


FIGURE 4: BOILING ACOUSTIC SPECTRUMS WITH DIFFERENT INTENSITIES OF WHITE NOISES.

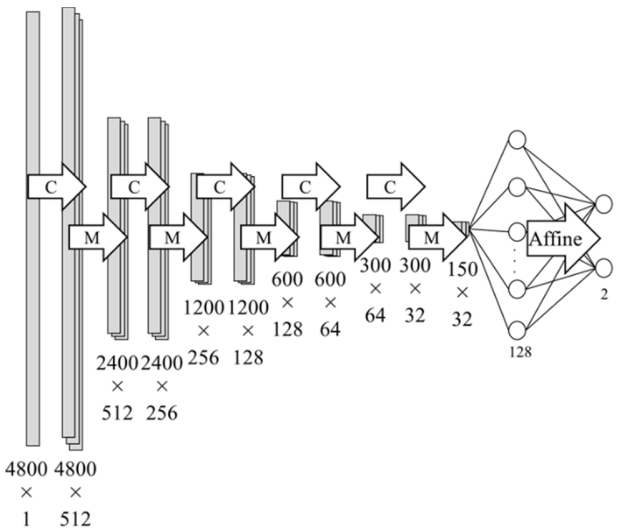
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We employ the CNN to detect the boiling inception. The network architecture is shown in FIGURE 5 (a). Character C denotes the convolution process, and character M does the max pooling. Node and kernel numbers are shown in the figure. The optimizer is Adam. The activation function for the output layer is the softmax function. That for the others is the ReLU function. The loss function is cross entropy. The other method we employ is to first perform denoising via AE, and then to use the CNN to detect the boiling inception. Hereafter we name it the AE-invoked CNN. the AE network architecture is shown in FIGURE 5 (b). The character U denotes upsampling. The dataset for the AE is dimensional. The timewise data dimension is four. The node and kernel numbers are also shown in the figure. The optimizer is Adam. The activation function is the ReLU function. The F-value is employed to quantitatively evaluate their classification performance. The F-value is calculated from the following equation:

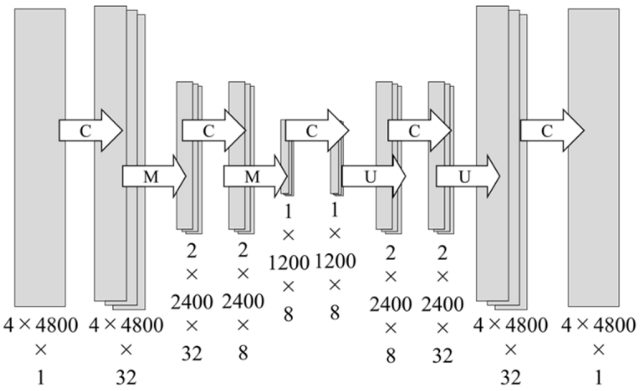
$$F = \frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{3}$$

The parameters of TP, TN, FP, and FN are defined as shown in TABLE 2.

We train and test each of the deep learning models ten times and evaluate their categorization performance. FIGURE 6 shows the F-value of both methods. The average value is shown with the maximum and minimum values. From the figure, we can see that, at a relatively low level of superposing white noises, CNN shows high accuracy performance in acoustic boiling sensing. On the other hand, at a relatively high level of noise where the noise/signal ratio is up to 9, the CNN classification together with the AE denoising of our novel method shows higher accuracy performance than that of single use of CNN. This result indicates that acoustic boiling sensing based on deep learning has a high potential to work even in acoustically noisy conditions.



(a) CNN.



(b) AUTOENCODER.

FIGURE 5: NEURAL NETWORK ARCHITECTURE.

TABLE 2: DEFINITION OF TP, TN, FP, FN.

Judgment result \ Actual condition	Non-Boiling	Boiling
Non-Boiling	TN	FN
Boiling	FP	TP

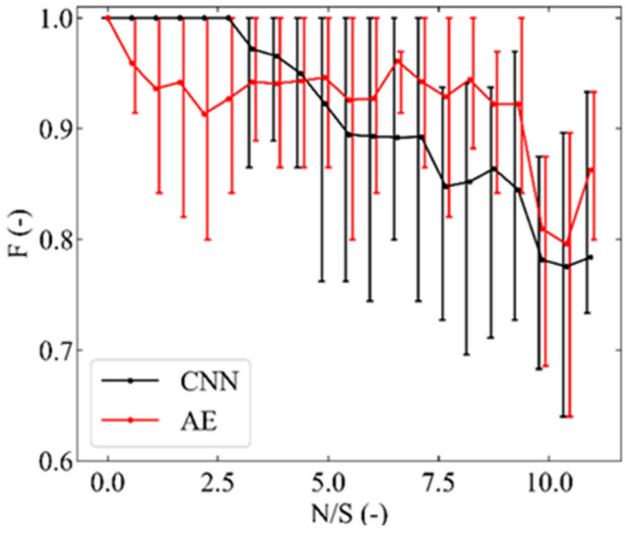


FIGURE 6: F-VALUE OF CNN AND AE-INVOKED CNN.

5. CONCLUSION

The acoustic boiling sensing based on deep learning has a high potential to work even in highly noisy conditions. The boiling acoustics have not been fully understood yet. In the present study, we find that some characteristics of the boiling

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acoustics are consistent with the resonance vibration of the heating wire. This finding contributes to elucidating the physics of boiling acoustics. In addition, it helps us detect boiling occurrences with high accuracy judging from the acoustic characteristics' patterns.

One of remaining issues before a practical use of the present method is to characterize and comprehend boiling acoustics of the sodium, although the present study indicates that the resonance vibration of the heating body can explain a part of the boiling acoustics. The issue needs to be done as our future work.

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